The Alaoglu–Erdős Integer-Exponent Conjecture

Machine-Checked Partial Results,

Exact Conditional Structure,

and Research Directions

A combined survey and research report on the corpus in

https://github.com/VladimirReshetnikov/ProveIt

Bottom line. No unconditional proof of the two-base conjecture is obtained, and the reduction to the four exponentials conjecture explains why none should be expected from routine combination of the present methods. The combined corpus does establish, in kernel-verified Lean: a zero-free region $2^x < 4096$ (any nonintegral solution has $x \geq 12$); transcendence of every hypothetical nonintegral solution; a canonical primitive odd core; a unique least nonintegral solution β whose output pair is affinely primitive; and the exact conditional rigidity theorem

$$\mathcal{S} = \{n + k\beta : n, k \in \mathbb{N}_0\},$$

together with logarithmic counting, a linear dyadic-block bound, sharp arbitrary-order progression obstructions, density zero, explicit Diophantine lower bounds, the 3-smooth classification, and the equivalence with both fractional-part gaps and a localized-radical exclusion. Exact unit-block counts, the sharp quadratic cumulative asymptotic, and block averages for every finite trigonometric polynomial are kernel-verified. The stronger continuous-test-function and interval-counting forms of blockwise Weyl equidistribution are proved here only at the paper level. An independent interval computation excludes all nonintegral candidates with $2^x < 2^{20}$; that computation is explicitly *not* a Lean kernel proof. The irreducible remainder is the open radical assertion: no odd $u > 1$ has $u^{\log_2 3} \in \mathbb{Z}[1/3]$.

This document merges, updates, and extends two earlier AI-assisted reports

(Claude Fable 5, Anthropic; GPT-5.6 Pro, OpenAI), against the repository state of

the combined-report base `83ef82ab1` and the kernel-checked feature modules listed in Appendix B.
All Lean identifiers cited below are the authoritative statements.

August 20, 2026

Abstract

The Alaoglu–Erdős conjecture asks whether a real x satisfying $2^x, 3^x \in \mathbb{Z}$ must be an integer. It is a particularly concrete unresolved instance of the four exponentials problem: the three-base analogue with $5^x \in \mathbb{Z}$ added is a theorem, formalized in the repository by an interpolation-determinant argument, while the two-base statement sits exactly at the open 2×2 transcendence threshold.

This report is the merged and expanded successor of two documents: a survey of the machine-checked partial results and an independent research report containing an audit and new conditional structure theory. It serves four purposes. First, it catalogues the kernel-verified Lean corpus — zero-free regions through $2^x < 4096$, Gelfond–Schneider transcendence restrictions, multiplicative rank at most two, a canonical primitive odd core, an exact conditional solution monoid with least generator, simultaneous primitive normalizations of its two outputs, a denominator-controlled rational-third-base extension, sharp arbitrary-order progression obstructions, zero density, shrinking-target Diophantine bounds, the 3-smooth classification of auxiliary bases, the four-exponentials reduction, and unconditional equivalences with both fractional-part gaps and a localized-radical exclusion — with the precise Lean identifiers and their verification status. Second, it records the now kernel-verified conditional structure: if a counterexample exists, a canonical primitive odd core w, rational-power index d, and least nonintegral solution $\beta = a + d \log_2 w$ exist, and the full solution set is the free additive monoid $\mathbb{N}_0 + \mathbb{N}_0\beta$; exact dyadic-block counts, their sharp quadratic asymptotic, and finite-trigonometric-polynomial block averages are kernel-verified. Full equidistribution for arbitrary continuous tests or interval indicators remains a paper corollary. Third, it records the now kernel-verified general higher-difference obstruction for arbitrary progression length and an independent 40-digit interval verification excluding all nonintegral candidates below 2^{20} under a stated software trust boundary. Fourth, it consolidates the research program: the conjecture is kernel-verified equivalent to the radical statement that $u^{\log_2 3} \notin \mathbb{Z}[1/3]$ for every odd $u > 1$. Primitive-base reduction, exact radical degree, and the uniform finite-difference API now have kernel-verified forms; the radical exclusion itself remains the open boundary.

Contents

1 Executive summary

1.1 What is proved, and at what level of trust

The full conjecture remains open. Several routes were pushed to their exact failure points: the three-base determinant method does not shrink to two bases without proving a case of the four exponentials conjecture; equidistribution of multiples does not defeat exponentially shrinking integrality gaps; local curvature obstructions cannot reach the sparse global structure; and every successful reduction terminates at the same transcendence-theoretic wall, now isolated in Conjecture 6.3.

To keep unlike kinds of evidence separate, every important statement in this report carries one of four labels.

[kernel-verified Lean] A proof source is present in the repository, lies on the audited `ExponentialIdentities.lean` import surface, and has been accepted by the Lean kernel in direct or full builds of the cited modules (toolchain 4.31/4.32 series, Mathlib pinned by the repository manifest). No `sorry`, no extra axioms, no `native_decide`.

[paper proof in this report] A conventional mathematical proof is supplied in this document. It is intended to be suitable for formalization, but no claim of kernel checking is made.

[interval computation] A finite statement was checked by outward-rounded interval arithmetic. The script, parameters, worst margins, and manifest hashes are recorded in Section 8 and Appendix D. The trusted base includes CPython and `mpmath`; this is not interchangeable with a Lean certificate.

[classical theorem] A classical theorem quoted from the literature.

An earlier audit, performed against commit `423ac24`, correctly observed that three modules cited by the survey (`FiniteCheck4096`, `OddCore`, `FractionalPartDensity`) were absent from that commit tree. That discrepancy is resolved: the three modules were subsequently committed in `98646b455` and are kernel-verified; the survey's temporary pending-verification remark was removed in `716617d72`. Statements from these modules are therefore labeled **[kernel-verified Lean]** below.

The current feature wave adds directly kernel-checked modules `PrimitiveOddCore`, `LeastSolution`, `ThreeDenominatorNormalization`, `RationalPowerIndex`, `PrimitiveGenerator`, `ExactCounting`, `QuadraticAsymptotic`, `BlockWeyl`, `BlockEquidistribution`, `OutputNormalization`, and `RationalThirdBase`. The radical modules are `RadicalReformulation`, `PrimitiveRadical`, and `RadicalDegree`; the progression modules are `SharperProgressions`, `ThirdDifference`, and `HigherDifference`. Their public headline theorems were audited with `##print axioms`; only the standard Mathlib axioms `propext`, `Classical.choice`, and `Quot.sound` occur.

1.2 Headline results

1. **Verified zero-free region.** If $2^x, 3^x \in \mathbb{Z}$ and $2^x < 4096$, then $x \in \mathbb{Z}$; equivalently every nonintegral solution has $x \geq 12$. **[kernel-verified Lean]** An independent interval scan extends the finite barrier to $2^x < 2^{20}$, i.e. $x > 20$. **[interval computation]**
2. **Transcendence.** Every nonintegral solution is transcendental; for x with $2^x \in \mathbb{Z}$, algebraicity of x is *equivalent* to integrality. $\vartheta = \log_2 3$ is transcendental. **[kernel-verified Lean]**
3. **Structure.** Conditional on a counterexample there is a unique gcd-normalized odd core $w > 1$, every candidate has unique coordinates $2^i w^j$, and the solution set has the exact free-monoid classification $\mathcal{S} = \mathbb{N}_0 + \mathbb{N}_0\beta$ with a unique least nonintegral generator. Candidate counts below N are at most $c\log_2 N \cdot c\log_3 N$, and each dyadic block $[2^T, 2^{T+1})$ holds at most $c\log_3(2^{T+1})$ candidates. The exact block count $\lfloor (N+1)/\beta \rfloor$, the limit $C(T)/T^2 \to 1/(2\beta)$, and the equivalence between the conjecture and subquadratic growth along 2^T are kernel-verified. Exact block averages converge to the constant coefficient for every finite trigonometric polynomial. **[kernel-verified Lean]** The extension to all continuous tests and interval indicators remains paper-level. **[paper proof in this report]**
4. **Unconditional gap equivalences.** The conjecture is equivalent to the existence of any nonempty open subinterval of $(0,1)$ free of fractional parts of nonintegral solutions, and to fractional parts being bounded away from 0, or from 1. **[kernel-verified Lean]**
5. **Sharper local combinatorics.** For every $r \geq 2$, the sharp criterion $|\vartheta(\vartheta-1)\cdots(\vartheta-r+1)|h^r < n^{r-\vartheta}$ forbids $r+1$ candidates in arithmetic progression. The underlying arbitrary-order forward-difference mean-value identity is formalized as well, and for the three-term case a rational effective criterion $h^{400} \leq n^{83}$ (step exponent $83/400 = 0.2075$) is verified independently. **[kernel-verified Lean]**
6. **The irreducible remainder.** The conjecture is equivalent to: no odd integer $u > 1$ satisfies $u^\vartheta \in \mathbb{Z}[1/3]$, with a symmetric base-swapped formulation and an equivalent reduction to canonical power-primitive odd bases. The least radical index gives an irreducible binomial and the exact algebraic degree. **[kernel-verified Lean]** The exclusion itself remains open even for $u = 3$.

2 The problem and its equivalent forms

2.1 Original statement

Conjecture 2.1 (Alaoglu–Erdős, two-base integer form). *For every $x \in \mathbb{R}$,*

$$2^x \in \mathbb{Z} \quad \textit{and} \quad 3^x \in \mathbb{Z} \qquad \Longrightarrow \qquad x \in \mathbb{Z}.$$

Status: recorded in Lean as the proposition `AlaogluErdosConjecture`, without being asserted.

Since 2^x and 3^x are positive, the hypotheses actually put them in $\mathbb{N}$; and $2^x \in \mathbb{N}$ forces $x \geq 0$, since a negative exponent gives $0 < 2^x < 1$. All solutions therefore live in $[0, \infty)$ (`nonneg_of_two_rpow_integer`).

Questions of this type go back to the 1944 study of colossally abundant numbers by Alaoglu and Erdős [1], where the arithmetic nature of simultaneous rational powers p^x, q^x of distinct primes controls the quotients of consecutive colossally abundant numbers — a theme reaching back to Ramanujan's superior highly composite numbers [2]. The modern two-base integer form is elementary to state and is a direct special case of the four exponentials conjecture (Schneider's first problem [7]; see Waldschmidt [8, 11]). As of the dated literature check for this report, the four exponentials conjecture remains open.

2.2 Candidate-base normalization

Put

$$\vartheta := \frac{\log 3}{\log 2} = \log_2 3 = 1.584962500721156\ldots$$

and define the *candidate set*

$$\mathcal{C} := \{m \in \mathbb{N}_{>0} : m^\vartheta \in \mathbb{N}\}.$$

Lemma 2.2 (Exponent–candidate bijection)**.** *The maps $x \mapsto 2^x$ and $m \mapsto \log_2 m$ are mutually inverse bijections between $\mathcal{S} := \{x \in \mathbb{R} : 2^x, 3^x \in \mathbb{Z}\}$ and $\mathcal{C}$. Under the bijection, $3^x = m^\vartheta$.*

Proof. If $x \in \mathcal{S}$, set $m = 2^x \in \mathbb{N}$. Then $m^\vartheta = \exp(\log m \cdot \log 3/\log 2) = \exp(x \log 3) = 3^x \in \mathbb{N}$. Conversely, if $m \in \mathcal{C}$ and $x = \log_2 m$, then $2^x = m$ and the same identity gives $3^x = m^\vartheta \in \mathbb{N}$. ☐

Status: [kernel-verified Lean] —
`exists_twoBaseNaturalCandidate_of_two_three_rpow_integer`,
`TwoBaseNaturalCandidate.integer_powers`.

Integral solutions correspond exactly to powers of two (`twoBaseCandidateExponent_isInteger_iff_isPowerOfTwo`), so the conjecture is equivalent to

$$\mathcal{C} = \{2^n : n \in \mathbb{N}_0\}$$

(`alaogluErdosConjecture_iff_candidates_are_powers_of_two`). The candidate formulation is well suited to unique factorization, finite certification, finite differences of t^ϑ, and multiplicative-rank arguments, and it is the formulation in which most of the corpus is stated.

Two closure properties of $\mathcal{S}$ are elementary but essential: $\mathcal{S}$ is closed under addition and under multiplication by positive naturals, so it is an additive submonoid of $[0, \infty)$ containing $\mathbb{N}$ (`rpow_integer_add`, `rpow_integer_nat_mul`).

2.3 The four-exponentials reduction

Suppose x is nonintegral and $2^x, 3^x$ are algebraic. The rational-exponent argument (Lemma 2.4) shows that x cannot be rational, so $1, x$ are $\mathbb{Q}$-linearly independent; and $\log 2, \log 3$ are $\mathbb{Q}$-linearly independent since $2^a 3^b = 1$ forces $a = b = 0$. The four exponentials built from these two pairs are

$$e^{1\cdot\log 2} = 2, \quad e^{1\cdot\log 3} = 3, \quad e^{x\log 2} = 2^x, \quad e^{x\log 3} = 3^x,$$

all four algebraic — contrary to the four exponentials conjecture. This reduction is formalized: `alaogluErdosConjecture_of_realFourExponentialsConjecture`. [kernel-verified Lean]

The reduction is a diagnostic, not just context: any proposed elementary argument must exploit integrality, positivity, or arithmetic structure beyond bare algebraicity of those four values, or else it would prove a substantial case of the four exponentials conjecture by another name. By contrast, three multiplicatively independent bases fall to the six exponentials mechanism, and the special case needed here is fully formalized:

Theorem 2.3 (Three-base theorem). *If $2^x, 3^x, 5^x \in \mathbb{Z}$ for a real x, then $x \in \mathbb{Z}$.*

Status: [kernel-verified Lean] — modules `IntegerExponent`, `IntegerExponent.DeterminantBound`, `IntegerExponent.ExponentialKernel`, `IntegerExponent.Grid`, following Waldschmidt's square-determinant proof [9].

In outline, for irrational x one forms a large square matrix $\mathcal{A}_{ij} = \exp(a_i b_j)$, where row arguments encode monomials in $2, 3, 5$ and column arguments encode affine forms in $1, x$. Irrationality makes the column parameters pairwise distinct, multiplicative independence the row parameters; a total-positivity argument gives $\det \mathcal{A} \neq 0$; integrality of all entries forces $|\det \mathcal{A}| \geq 1$; and an interpolation estimate with explicit size choices gives $|\det \mathcal{A}| < 1$, a contradiction. The dimension count is critical: with three independent row directions and two column directions the analytic bound beats the arithmetic bound, while in the 2×2 case the same balance lands exactly on the classical boundary where decay below one fails. A successful two-base determinant would be a new four-exponentials argument.

A denominator-controlled rational third output. The rational-output extension now formalized is deliberately conditional. Let $a > 1$ and assume the monomial map

$$(i, j, k) \longmapsto 2^i 3^j a^k$$

is injective. If $2^x = m_2$ and $3^x = m_3$ are natural numbers, $a^x = q \in \mathbb{Q}$, and for some $u, v \in \mathbb{N}_0$ the reduced denominator satisfies

$$\mathrm{den}(q) \mid m_2^u m_3^v,$$

then $x \in \mathbb{Q}$, and in fact $x \in \mathbb{Z}$ because 2^x is integral. A concrete corollary obtains the monomial injectivity when a has a prime divisor different from 2 and 3.

On the determinant side, a nonzero $m \times m$ real determinant whose entries become integers after multiplication by a common positive denominator Q satisfies $|\det A| \geq Q^{-m}$. The explicit analytic estimate also retains a denominator margin: for $m = 2r$, $r > 2$, $192C \leq r$, and $bs + 2r < r^2$, the verified upper-bound expression remains strictly below 1 even after multiplication by b^s.

Status: [kernel-verified Lean] — `one_div_pow_le_abs_det_of_common_denominator`, `exponential_det_numeric_bound_with_denominator`, `rational_of_two_three_a_rpow_rational_of_den_dvd_outputs`, and `integer_of_two_three_a_rpow_rational_of_prime_factor_of_den_dvd_outputs` in `RationalThirdBase`. These are denominator-divisibility and explicit-margin results; the unrestricted determinant grid for an arbitrary rational third-base output has *not* been completed.

2.4 Rational and algebraic exponents

Lemma 2.4 (Rational exponent)**.** *If $x \in \mathbb{Q}$ and $2^x \in \mathbb{Z}$, then $x \in \mathbb{Z}$. Consequently every nonintegral solution is irrational.*

Proof. Write $x = a/b$ in lowest terms with $b > 0$. If $2^{a/b} = m \in \mathbb{N}$, then $2^a = m^b$; comparing the exponent of 2 gives $a = b\, v_2(m)$, so $b \mid a$ and $b = 1$. $\square$

Status: **[kernel-verified Lean]** —
`IntegerExponent.integer_of_rational_of_two_rpow_integer`,
`irrational_of_not_integer_of_two_rpow_integer`.

Gelfond–Schneider upgrades irrational to transcendental; see section 4.6.

3 Repository timeline and verification status

The formal development advanced in three waves plus documentation. All commits are in the repository [13]; the mathematical role of each is summarized below.

Commit	Role
`f492c18f1`	Three-base theorem via interpolation determinants (`IntegerExponent` and submodules).
`5ed4e6185`	Merge of the three-base branch.
`a37d97f9d`	Basic two-base module: the conjecture as a proposition, closure under natural multiples, irrationality of nonintegral solutions, endpoint gaps, numerical bounds for ϑ, the $2^x < 10$ exclusion, second-difference obstruction, aligned-block 2/3 counting.
`4a9b03d12`	Major corpus: finite checks through $2^x < 256$; arithmetic descent; multiplicative rank; candidate structure; zero density; shrinking targets; the 3-smooth theorem; the four-exponentials reduction; transcendence consequences; and a complete Gelfond–Schneider port.
`18a85e6b8`	Merge of the two-base branch.
`423ac24f0`	First survey document (`two-base-partial-results.tex`) with rendered PDF.
`98646b455`	Three further modules, kernel-verified: `FractionalPartDensity` (density dichotomy and gap equivalences), `OddCore` (common odd core, improved counting, dyadic blocks), `FiniteCheck4096` (zero-free region to $2^x < 4096$).
`716617d72`	Survey finalized after kernel acceptance of the 4096 region.
`11ac3093f`	Independent research report (audit at `423ac24f0`, exact conditional structure, interval scan to 2^{20}); one of the two sources merged into the present document.

Commit	Role
current tree	Kernel-checked feature modules formalizing the primitive common odd core, least-solution primitivity, simultaneous output normalization, power-of-three denominator normalization, rational-power index and exact primitive generator, exact conditional counting and its quadratic asymptotic, finite-Fourier block averages, denominator-controlled rational third-base outputs, primitive localized-radical reduction and exact radical degree, and uniform higher finite differences.

The audit performed in `11ac3093f` treated results of the then-uncommitted modules as article-level claims; with `98646b455` in the tree and kernel-verified, those results are now cited as committed Lean. The interval scan to 2^{20} remains at the **[interval computation]** level by design. The later feature modules listed in the final row prove the cumulative asymptotic and finite-trigonometric-polynomial block averages. They do not yet formalize the stronger continuous-test-function or interval-counting equidistribution theorem; that distinction is maintained throughout this report.

4 The kernel-verified corpus

Throughout this section x denotes a hypothetical nonintegral solution, $m = 2^x \in \mathbb{N}$, $n = 3^x \in \mathbb{N}$. Every statement here is **[kernel-verified Lean]** unless labeled otherwise.

4.1 Fractional-part gaps and shrinking targets

An integral power caught strictly between consecutive integral powers of its base must clear both by at least one full unit; dividing through normalizes this to the fractional part.

Theorem 4.1 (Fractional-part gaps)**.** *Let $x \in (N, N+1)$ with $N \in \mathbb{N}$, $t = x - N$, and suppose $2^x, 3^x \in \mathbb{Z}$. Then*

$$1 + 2^{-N} \le 2^t \le 2 - 2^{-N}, \qquad 1 + 3^{-N} \le 3^t \le 3 - 3^{-N}.$$

Status: **[kernel-verified Lean]** — `two_three_fractional_part_gaps`, `two_three_rpow_gaps_near_nat`, from `rpow_integer_gap_between_consecutive_powers`.

Near the endpoints the excluded zones have width $\asymp 2^{-N}$: they *shrink* as the integer part grows. Applied to all multiples kx (which are again solutions), this yields explicit Diophantine lower bounds.

Theorem 4.2 (Effective irrationality bound)**.** *Let $x \notin \mathbb{Z}$ satisfy $2^x \in \mathbb{Z}$ and write $m = 2^x$. Then for all integers p and all $k \ge 1$,*

$$\left|x - \frac{p}{k}\right| \ \ge\ \frac{1}{2\,k\,m^k}.$$

Exact (slightly stronger) radii, and two-base versions taking the better of the base-2 and base-3 radii, are formalized as well.

Status: [kernel-verified Lean] — `rational_approximation_exponential_lower_bound`, `two_three_multiple_distance_to_integer_lower_bound`, `exists_natural_base_rational_approximation_lower_bound` in `ShrinkingTarget`.

In shorthand, $\|kx\|_{\mathbb{Z}} \gtrsim 2^{-kx}$. This bound is compatible with ordinary Diophantine approximation: Dirichlet guarantees infinitely many k with $\|kx\|_{\mathbb{Z}} < 1/k$, and the shrinking radius 2^{-kx} is incomparably smaller. The interplay between this exponential gap and equidistribution is discussed in section 9.

The bound dualizes into a growth constraint on the continued fraction of a solution: if $|x - p/q| < 1/(qQ)$ with $q, Q \geq 1$, then $Q < 2m^q$. Applied to consecutive convergent denominators, which satisfy $|x - p_n/q_n| < 1/(q_n q_{n+1})$, this gives

$$q_{n+1} \; < \; 2\, m^{\,q_n} \, :$$

the partial quotients of a hypothetical nonintegral solution are capped by a single exponential in the preceding denominator, with base its own output $m = 2^x$.

Status: [kernel-verified Lean] — `approximation_quality_lt_of_noninteger_solution` in `DenominatorGrowth`.

4.2 Verified zero-free regions and certificate architecture

Theorem 4.3 (Zero-free region)**.** *If $2^x, 3^x \in \mathbb{Z}$ and $2^x < 4096$, then $x \in \mathbb{Z}$. Equivalently, every nonintegral solution satisfies $x \geq 12$, and every candidate that is not a power of two is at least* 4096*.*

Status: [kernel-verified Lean] — `integer_of_two_three_rpow_integer_of_two_rpow_lt_4096`, `twelve_le_of_not_integer_of_two_three_rpow_integer`, `le_of_twoBaseNonintegerCandidate_4096` in `FiniteCheck4096`, extending `FiniteCheck` ($2^x < 100$) and `FiniteCheck256` ($2^x < 256$, hence $x \geq 8$).

The certificate architecture is worth recording, both because it is the repository's model of a trustworthy large computation and because scaling it is a named research direction. For each non-power-of-two m the certificate is a pair of exact integer power comparisons against rational enclosures $L_p/L_q < \vartheta < U_p/U_q$ drawn from continued-fraction convergents of ϑ: from

$$a^{L_q} < m^{L_p} \quad \text{and} \quad m^{U_p} < (a+1)^{U_q}$$

one traps $a < m^{\vartheta} < a+1$, so $m^{\vartheta} \notin \mathbb{Z}$. The $2^x < 256$ stage verifies per-value certificates by `norm_num` with three enclosure tiers (exponent sizes $\sim$500, $\sim$1500, $\sim$25000). The extension to 4096 verifies 3836 certificates through a single kernel-replayed `decide` over a table generated by PARI/GP and revalidated independently with exact big-integer arithmetic in Python. Five tiers suffice, with tier populations 78, 488, 3241, 28, 1:

tier	lower convergent L_p/L_q	upper convergent U_p/U_q
0	569/359	485/306
1	1054/665	1539/971
2	25781/16266	24727/15601
3	50508/31867	125743/79335
4	176251/111202	301994/190537

Only the value $m = 2762$, whose ϑ-power is within $3 \cdot 10^{-9}$ of an integer in logarithmic scale, needs tier 4; its certificate compares integers of roughly 90,000 digits, which the kernel's exact arithmetic replays without difficulty. No axioms beyond the Lean kernel are involved; in particular no `native_decide`.

4.3 Arithmetic descent

Theorem 4.4 (Affine descent). *If the two integral outputs factor simultaneously as $2^x = 2^r u^k$ and $3^x = 3^r v^k$ with $r \geq 0$, $k \geq 1$, then $(x - r)/k$ is again a solution. Combined with the $x \geq 8$ region this gives the unconditional constraint*

$$r + 8k \leq x$$

for nonintegral x; in particular common perfect-power shapes of the two outputs are strongly restricted.

Status: [kernel-verified Lean] — `affine_descent_integral_powers`, `base_depth_add_eight_mul_degree_le`, `min_output_base_factorization_add_eight_le` in `ArithmeticRigidity`. With the 4096 region the constant 8 improves to 12 by bookkeeping; with the interval scan of section 8 it would improve to 20 at the **[interval computation]** level.

For the *least* counterexample the same descent later becomes an exact primitivity statement (Corollary 5.6): no nontrivial matched decomposition exists at all.

4.4 Local progression obstructions and zero density

The function $t \mapsto t^\vartheta$ is strictly convex with second differences at moderate steps lying strictly between 0 and 1 — hence never integers.

Theorem 4.5 (Committed AP obstruction). *Let $n, h \geq 1$ with $h^5 \leq n$. Then $n, n + h, n + 2h$ are not all candidates. In particular no three consecutive naturals are candidates, and among the positive integers up to $3N$ at most $2N$ are candidates.*

Status: [kernel-verified Lean] — `not_three_arithmetic_progression_twoBaseNaturalCandidates`, `not_three_consecutive_logThreeDivLogTwo_rpow_integers`, `twoBaseNaturalCandidate_Icc_card_le`. The sharper criterion $\vartheta(\vartheta-1)h^2 < n^{2-\vartheta}$ is kernel-verified in `SharperProgressions`, and the analogous four-term criterion is kernel-verified in `ThirdDifference`. The arbitrary-order statement for every $r \geq 2$ is `not_arithmetic_progression_twoBaseNaturalCandidates_of_curvature` in `HigherDifference`.

The hypothesis bounding h is essential: for steps comparable to n the second difference is large and the obstruction vanishes. This locality is a genuine barrier — a *global* progression obstruction would, for example, forbid candidates with odd part $2^s \pm 1$ via progressions such as $(2^i,\ 2^{i+t-1},\ 2^i(2^t-1))$, but no such global statement is available; under a counterexample the exact monoid of section 5 shows sparse candidates whose additive relations the curvature argument cannot see.

Feeding the local obstruction into the theory of Roth numbers gives quantitative sparsity.

Theorem 4.6 (Zero density)**.** *For every $H \geq 1$ the number of candidates below N is at most $H^5 + \lceil (N-H^5)/H \rceil\, r_3(H)$, where $r_3(H)$ is the maximal size of a 3-AP-free subset of $\{0, \dots, H-1\}$. Consequently the candidate counting function is $o(N)$: the candidates, and a fortiori the values 2^x of nonintegral solutions, have asymptotic density zero.*

Status: [kernel-verified Lean] — `twoBaseNaturalCandidatesBelow_card_le`, `twoBaseNaturalCandidatesBelow_isLittleO_id`, `tendsto_twoBaseNonintegerCandidatesBelow_density_zero` in `ZeroDensity` and `NonintegerDensity`.

4.5 Multiplicative rank two and the common odd core

Theorem 4.7 (Rank two)**.** *The prime-exponent vectors $\mathbf{v}(a) = (v_p(a))_p$ of all candidates span a rational space of dimension at most two. Consequently at most $(\operatorname{clog}_2 N)^2$ candidates lie below N.*

Status: [kernel-verified Lean] — `finrank_span_primeExponentVectors_le_two`, `twoBaseNaturalCandidatesBelow_card_le_clog_sq` in `MultiplicativeRank`. The mechanism abstracts the determinant method: three candidates with independent exponent vectors would give three multiplicatively independent bases, contradicting Theorem 2.3 through the irrationality of ϑ (`irrational_logThreeDivLogTwo`).

Since 2 is itself a candidate, all odd parts lie on at most one further rational direction. Pairwise this yields the cross-valuation identity, and globally a common odd core.

Lemma 4.8 (Cross-valuation identity)**.** *Let $a, b \in \mathcal{C}$ and let p be an odd prime with $v_p(a) > 0$. Then for every odd prime q,*

$$v_p(a)\ v_q(b) = v_p(b)\ v_q(a)\,.$$

Status: [kernel-verified Lean] — `odd_factorization_cross_mul` in `CandidateStructure`; the fixed-power relations `fixed_power_relation_of_odd_prime_factor` follow.

Theorem 4.9 (Common odd core)**.** *There is a single odd $w > 1$ such that every candidate equals $2^i w^j$ for some $i, j \in \mathbb{N}_0$; candidates that are not powers of two have $j \geq 1$. In exponent coordinates: every solution has the form $x = i + j \log_2 w$ with $i, j \in \mathbb{N}_0$.*

Status: [kernel-verified Lean] — `exists_common_odd_core`, `exists_common_odd_core_split`, `exists_common_odd_core_exponent` in `OddCore`. The canonical (gcd-normalized) version with uniqueness of w and of each coordinate pair is `existsUnique_primitive_common_odd_core` in `PrimitiveOddCore`.

Corollary 4.10 (Improved counting)**.** *The number of candidates below N is at most $\mathrm{clog}_2 N \cdot \mathrm{clog}_3 N$, and each dyadic block $[2^T, 2^{T+1})$ contains at most $\mathrm{clog}_3(2^{T+1}) \approx 0.631\,(T+1)$ candidates. In exponent coordinates, the number of solutions with $x \in [T, T+1)$ grows at most linearly in T, against the quadratic global bound.*

Status: [kernel-verified Lean] — `twoBaseNaturalCandidatesBelow_card_le_clog_mul_clog`, `twoBaseNonintegerCandidatesBelow_card_le_clog_mul_clog`, `twoBaseNaturalCandidatesInDyadicBlock_card_le` in `OddCore`.

The logarithmic-square order cannot be improved by counting alone: if a candidate with an odd prime factor exists, multiplicative closure generates injective boxes $2^i a^j$, so the count below $2^k a^l$ is at least kl (`twoBaseNaturalCandidatesBelow_card_ge_box`, `twoBaseNonintegerCandidatesBelow_card_ge_box`) — a conditional lower bound of the same quadratic-in-logarithm shape, made exact in section 5.4.

4.6 Transcendence

The corpus contains a complete Lean proof of the Gelfond–Schneider theorem (Hilbert's seventh problem), following Hua's exposition [4], ported from Michail Karatarakis's Mathlib formalization (Mathlib PR #42911, Apache 2.0).

Theorem 4.11 (Gelfond–Schneider)**.** *If α, β are algebraic, $\alpha \neq 0, 1$, and β is irrational, then α^β is transcendental.*

Status: [kernel-verified Lean] — `GelfondSchneider.transcendental_cpow_of_isAlgebraic_of_irrational`.

Corollary 4.12 (Algebraic–integral dichotomy)**.** *If $2^x \in \mathbb{Z}$ then x is either an integer or transcendental; algebraicity of such x is equivalent to integrality. Every nonintegral two-base solution is transcendental, and $\vartheta = \log_2 3$ is transcendental.*

Status: [kernel-verified Lean] —
`transcendental_of_not_integer_of_two_rpow_integer`,
`isAlgebraic_iff_integer_of_two_rpow_integer`,
`transcendental_of_not_integer_of_two_three_rpow_integer`,
`transcendental_logThreeDivLogTwo`.

The dichotomy does not resolve the conjecture — the unknown exponent is allowed to be transcendental — but it constrains every construction below: the primitive generator β of section 5 is transcendental, as is $\log_2 w$ for the odd core.

4.7 Classification of auxiliary bases: the 3-smooth theorem

Theorem 4.13 (3-smooth classification)**.** *Suppose $x \notin \mathbb{Z}$ and $2^x, 3^x \in \mathbb{Z}$. Then for a positive natural a, the power a^x is an integer if and only if $a = 2^u 3^v$ is 3-smooth. In particular a third base with any prime factor other than 2 and 3 forces $x \in \mathbb{Z}$.*

Status: [kernel-verified Lean] —
`rpow_integer_iff_eq_two_pow_mul_three_pow_of_not_integer`,
`integer_of_two_three_a_rpow_integer_of_prime_factor` in `ThreeSmooth`.

This is an exact classification of bases for a *fixed* hypothetical counterexample. It should not be confused with the classification of candidate values 2^x as x ranges over all solutions, which is the subject of the next section. Note also the scope limitation explained in section 9: the theorem controls the quantity b^x, not divisibility of the integer 2^x by b.

4.8 Unconditional gap equivalences

Because natural multiples of a solution are solutions, one nonintegral solution generates infinitely many, with irrational generator; Dirichlet approximation and a rotation-stepping argument spread their fractional parts densely.

Theorem 4.14 (Density dichotomy)**.** *If some $x \notin \mathbb{Z}$ satisfies $2^x, 3^x \in \mathbb{Z}$, then for every $t \in [0,1]$ and $\varepsilon > 0$ there is a nonintegral solution y with $|\{y\} - t| < \varepsilon$; and the set of nonintegral solutions is infinite.*

Theorem 4.15 (Gap equivalences)**.** *Each of the following is equivalent to Conjecture 2.1:*

(i) *some nonempty open interval $(u, v) \subseteq (0,1)$ contains no fractional part of a nonintegral solution;*

(ii) *the fractional parts of nonintegral solutions are bounded away from 1;*

(iii) *the fractional parts of nonintegral solutions are bounded away from 0.*

Status: [kernel-verified Lean] — `dense_fract_of_noninteger_solution`, `infinite_noninteger_solutions_of_exists`, `alaogluErdosConjecture_iff_fract_avoids_interval`, `alaogluErdosConjecture_iff_fract_bounded_away_one`, `alaogluErdosConjecture_iff_fract_bounded_away_zero` in `FractionalPartDensity`, built on the reusable `exists_pos_nat_fract_close_of_irrational`.

Remark 4.16 (Consistency pincer). Theorem 4.1 bounds $\{x\}$ away from 0 and 1 by margins $\asymp 2^{-\lfloor x \rfloor}$ that decay with the height of the solution, while Theorem 4.14 produces fractional parts arbitrarily close to 0 and 1 along multiples of unbounded height. The two results meet edge-to-edge without contradiction, and Theorem 4.15 shows the decay is essential: *any* height-independent gap, however small and wherever located, is exactly as strong as the full conjecture. The elementary gap bounds are therefore of optimal shape, and the fractional-part axis measures the entire remaining distance to Conjecture 2.1.

These equivalences are *unconditional* biconditionals. Conditional on failure, the exact structure of the next section gives kernel-verified convergence for every finite trigonometric polynomial and the paper-level strengthening to full blockwise Weyl equidistribution (Theorem 5.10).

5 Exact conditional structure of all solutions

This section develops the exact rigidity theory of a hypothetical counterexample. The canonical primitive odd core, rational-power index, denominator normalization, unique least solution, simultaneous primitive output normalization, exact free-monoid classification, exact unit-block counts, the cumulative quadratic asymptotic, and finite-Fourier block averages are now **[kernel-verified Lean]**. Full blockwise Weyl equidistribution for arbitrary continuous tests or interval indicators remains **[paper proof in this report]**.

5.1 The primitive common odd core

Theorem 5.1 (Primitive common odd core)**.** *Assume $\mathcal{C}$ contains a non-power of two. Then there is a unique odd integer $w > 1$ that is power-free in the sense*

$$\gcd\{v_p(w) : p \mid w\} = 1,$$

and every candidate $b \in \mathcal{C}$ has a unique representation

$$b = 2^i w^j, \qquad i, j \in \mathbb{N}_0.$$

Proof. Choose a non-power-of-two candidate a. Its odd valuation vector $A = (A_q)_{q \neq 2}$, $A_q = v_q(a)$, is nonzero and finitely supported. Let

$$g = \gcd\{A_q : A_q > 0\}, \qquad c_q = A_q/g, \qquad w = \prod_{q \neq 2} q^{c_q}.$$

Then w is odd, greater than one, and the positive c_q have gcd one.

Let $b \in \mathcal{C}$ and set $B_q = v_q(b)$. Fix an odd prime p with $A_p > 0$. By Lemma 4.8, $A_p B_q = B_p A_q$ for all odd q. If $A_q = 0$, this forces $B_q = 0$. On the support of A it says

$$B_q = \frac{B_p}{A_p} A_q = \Big(\frac{gB_p}{A_p}\Big) c_q.$$

Put $t = gB_p/A_p \in \mathbb{Q}_{\geq 0}$. Every tc_q is an integer. Since the c_q have gcd one, there are integers r_q, finitely many nonzero, with $\sum r_q c_q = 1$; therefore $t = \sum_q r_q(tc_q) \in \mathbb{Z}$. Writing $j = t \in \mathbb{N}_0$, the odd part of b is exactly w^j, so $b = 2^i w^j$ with $i = v_2(b)$.

Uniqueness of (i, j) for fixed w follows from odd valuations and the 2-adic valuation. If w' is another primitive common core, the odd valuation vectors of w and w' generate the same rational ray and are both primitive integer vectors, hence equal. □

Status: [kernel-verified Lean] — `existsUnique_primitive_common_odd_core` in `PrimitiveOddCore`. The module defines `oddPrimeValuationGCD`, proves its equivalence with power-theoretic primitivity for $w > 1$, and kernel-checks both uniqueness of the normalized core and uniqueness of every pair (i, j).

The gcd normalization is what makes w canonical: it prevents hidden perfect powers from being absorbed into the exponent. Thus the earlier existence theorem for an arbitrary common core and the normalization argument above now both have kernel-verified counterparts.

Set

$$\alpha := \log_2 w, \qquad E := 3^\alpha = w^\vartheta.$$

Every candidate exponent then has the form $x = i + j\alpha$. The number α is irrational (else the rational-exponent lemma applied to $2^\alpha = w$ would make it integral, forcing the odd $w > 1$ to be a power of two), and indeed transcendental by Corollary 4.12. The candidate condition becomes

$$3^{i+j\alpha} = 3^i E^j \in \mathbb{N},$$

and at this stage the admissible pairs (i, j) still form an unknown subset of $\mathbb{N}_0^2$. The next subsection shows this subset is completely rigid.

5.2 The rational-power index and the least generator

Because a non-power-of-two candidate exists, some $j > 0$ has $E^j \in \mathbb{Q}$. Let

$$d := \min\{j \in \mathbb{N}_{>0} : E^j \in \mathbb{Q}\}.$$

Lemma 5.2 (All rational powers are multiples of the least one)**.** *For every $j \in \mathbb{Z}$: $E^j \in \mathbb{Q} \iff d \mid j$.*

Proof. $J := \{j \in \mathbb{Z} : E^j \in \mathbb{Q}\}$ is an additive subgroup of $\mathbb{Z}$ ($0 \in J$; products add exponents; reciprocals negate them), and every nonzero subgroup of $\mathbb{Z}$ is $d\mathbb{Z}$ for its least positive element. □

Status: [kernel-verified Lean] — `rationalPowerIndices`, `AddSubgroup.Int.exists_least_positive_generator`, and `exists_rationalPowerIndex` in `RationalPowerIndex`.

Write the positive rational E^d in lowest terms. Its denominator has no prime other than 3: taking a candidate with odd-core exponent $j = dk > 0$ and 2-exponent i, integrality of $3^i(E^d)^k$ shows any other prime in the reduced denominator would survive in the k-th power. Hence there are unique integers $a \geq 0$, $A \geq 1$ with

$$E^d = \frac{A}{3^a}, \tag{1}$$

and $3 \nmid A$ when $a > 0$. Define

$$\beta := a + d\alpha = a + d\log_2 w, \qquad M := 2^\beta = 2^a w^d, \qquad A = 3^\beta = 3^a E^d. \tag{2}$$

Thus β is itself a solution, irrational because α is.

Status: [kernel-verified Lean] — the denominator-support step is `exists_three_pow_denominator_of_rational_power` in `ThreeDenominatorNormalization`; the normalized integrality criterion is `three_pow_mul_normalized_pow_integer_iff` in `RationalPowerIndex`.

Theorem 5.3 (Exact primitive-generator theorem)**.** *Assume Conjecture 2.1 is false. With w, d, a, β, M, A as above:*

(i) *every solution has a unique representation $x = n + k\beta$ with $n, k \in \mathbb{N}_0$;*

(ii) *every candidate has a unique representation $m = 2^n M^k$ with $n, k \in \mathbb{N}_0$;*

(iii) *every output pair has a unique representation $(2^x, 3^x) = (2^n M^k, 3^n A^k)$;*

(iv) *β is the unique least nonintegral solution.*

Proof. Let $x \in \mathcal{S}$. By Theorem 5.1, $2^x = 2^i w^j$ for unique $i, j \in \mathbb{N}_0$, so $x = i + j\alpha$. From $3^x = 3^i E^j \in \mathbb{N}$ we get $E^j \in \mathbb{Q}$, so Lemma 5.2 gives $j = dk$. Using (1),

$$3^x = 3^i (E^d)^k = 3^{\,i-ak} A^k.$$

Because $A/3^a$ is reduced, this is an integer exactly when $i \geq ak$. Put $n = i - ak \in \mathbb{N}_0$; then

$$x = i + dk\alpha = n + k(a + d\alpha) = n + k\beta.$$

Conversely, for any $n, k \in \mathbb{N}_0$,

$$2^{\,n+k\beta} = 2^n M^k \in \mathbb{N}, \qquad 3^{\,n+k\beta} = 3^n A^k \in \mathbb{N},$$

so $n + k\beta \in \mathcal{S}$. If $n + k\beta = n' + k'\beta$ with $k \neq k'$, then $\beta = (n' - n)/(k - k') \in \mathbb{Q}$, a contradiction; hence the representation is unique. This proves (i), and (ii)–(iii) follow by exponentiation. A nonintegral solution has $k \geq 1$; the least value is attained at $(n, k) = (0, 1)$ and equals β, uniquely so by uniqueness of representation. □

Status: [kernel-verified Lean] — `exists_canonical_primitiveGenerator_of_not_alaogluErdosConjecture` in `PrimitiveGenerator`, built from `exists_exact_solution_monoid_of_fixed_common_oddCore` in `RationalPowerIndex`. Conditional on failure, $\mathcal{S}$ is not merely contained in a rank-two additive set: every solution has a unique pair of coordinates in the free additive monoid on 1 and the irrational least generator β. The set of occurring odd-core exponents is exactly $d\,\mathbb{N}_0$, with the power of two governed by one linear threshold.

Packaged algebraically: $(\mathbb{N}_0^2,+) \xrightarrow{\sim} (\mathcal{S},+)$ via $(n,k) \mapsto n+k\beta$, and $(\mathbb{N}_0^2,+) \xrightarrow{\sim} (\mathcal{C},\cdot)$ via $(n,k) \mapsto 2^n M^k$; the output-pair monoid $\mathcal{P}$ is freely generated by $(2,3)$ and (M,A). Every multiplicative relation among candidates is the obvious coordinate relation. This exact conditional model is the benchmark against which any future density, progression, or descent argument should be tested.

5.3 Primitivity of the least output pair

Leastness of β upgrades the repository's descent inequalities to exact exclusions.

Corollary 5.4 (No matched base depth)**.** *For the primitive pair $(M,A) = (2^\beta, 3^\beta)$: $\min\{v_2(M), v_3(A)\} = 0$; that is, M is odd or $3 \nmid A$ (possibly both).*

Proof. If both valuations were positive, then $2^{\beta-1} = M/2$ and $3^{\beta-1} = A/3$ would be integers, so $\beta - 1$ would be a smaller nonintegral solution. □

Corollary 5.5 (No common perfect-power degree)**.** *There is no $k \geq 2$ for which both M and A are perfect k-th powers.*

Proof. If $M = U^k$, $A = V^k$, then $2^{\beta/k} = U$ and $3^{\beta/k} = V$, and β/k is a smaller nonintegral solution. □

Corollary 5.6 (Exact affine primitivity)**.** *If $M = 2^r U^k$ and $A = 3^r V^k$ with $r \in \mathbb{N}_0$, $k \in \mathbb{N}_{>0}$, $U, V \in \mathbb{N}$, then $r = 0$ and $k = 1$.*

Proof. Affine descent (Theorem 4.4) gives the solution $(\beta - r)/k$, nonintegral since β is irrational; if $r > 0$ or $k > 1$ it is strictly smaller than β. □

Status: [kernel-verified Lean] — `exists_least_twoBaseNonintegerSolution`, `IsLeastTwoBaseNonintegerSolution.unique`, `IsLeastTwoBaseNonintegerSolution.not_both_base_dvd_outputs`, `IsLeastTwoBaseNonintegerSolution.no_common_perfect_power`, and `IsLeastTwoBaseNonintegerSolution.affine_primitive` in `LeastSolution`.

Theorem 5.7 (Simultaneous primitive output normalization)**.** *If the conjecture fails, its least nonintegral solution β admits simultaneous factorizations*

$$2^\beta = 2^a w^d, \qquad 3^\beta = 3^b v^e,$$

where $w > 1$ is odd and power-primitive, $v > 1$ is power-primitive with $3 \nmid v$, and $d, e > 0$. Moreover $a = 0$ or $b = 0$; a and b are the exact base-adic depths; d and e are the valuation gcds of the corresponding base-free parts; and

$$\gcd(\gcd(d,e), |b-a|) = 1.$$

Since every counterexample has $\beta \geq 12$, these normalizations also satisfy the size dichotomy

$$w^d \geq 4096 \qquad \text{or} \qquad v^e \geq 3^{12}.$$

Status: [kernel-verified Lean] — `exists_simultaneous_primitive_output_normalization`, `IsLeastTwoBaseNonintegerSolution.simultaneousCoordinateGCD_eq_one`, and `exists_large_primitiveCore_of_not_alaogluErdosConjecture` in `OutputNormalization`. The gcd-one statement simultaneously controls the two outer power degrees and the difference of the two base-adic depths; it is stronger than merely excluding a common perfect-power degree.

5.4 Exact counting

Theorem 5.8 (Exact unit-block count). *Assume a counterexample exists and let β be the primitive generator. For every $N \in \mathbb{N}_0$, the nonintegral solutions in $[N, N+1)$ are precisely*

$$x_{N,k} = \lceil N - k\beta \rceil + k\beta, \qquad 1 \le k \le \Big\lfloor \frac{N+1}{\beta} \Big\rfloor,$$

so their number is exactly $\lfloor (N+1)/\beta \rfloor$, and

$$\#(\mathcal{C} \cap [2^N, 2^{N+1})) = 1 + \Big\lfloor \frac{N+1}{\beta} \Big\rfloor.$$

Proof. For fixed $k \ge 1$ there is at most one integer n with $N \le n + k\beta < N+1$; such $n \ge 0$ exists iff $k\beta < N+1$, and then $n = \lceil N - k\beta \rceil$. Irrationality of β makes $(N+1)/\beta$ non-integral, giving the stated range; the one integral solution in the block is N. □

Status: [kernel-verified Lean] — `ExactCounting` proves the explicit ceiling-image parametrization, `twoBaseNonintegerSolutionsInUnitBlock_ncard`, `twoBaseNaturalCandidatesInDyadicBlock_card`, and the exact finite-sum identity `cumulativeTwoBaseCandidateCount_eq`. `QuadraticAsymptotic` identifies that cumulative count exactly with the candidate count below 2^T and proves the sharp bounds and limit in (3).

With the verified bound $\beta > 12$ the block count is at most $\lfloor N/12 \rfloor$ (and at most $\lfloor N/20 \rfloor$ under the interval scan). Summing over blocks, the cumulative candidate count below 2^T is

$$C(T) = T + \sum_{r=1}^{T} \Big\lfloor \frac{r}{\beta} \Big\rfloor = \frac{T^2}{2\beta} + O(T). \tag{3}$$

More precisely, Lean proves

$$\frac{T(T+1)}{2\beta} \le C(T) \le \frac{T(T+1)}{2\beta} + T, \qquad \frac{C(T)}{T^2} \longrightarrow \frac{1}{2\beta}.$$

This is conditionally sharper than the verified universal bounds ($(\mathrm{clog}_2 N)^2$ and $\mathrm{clog}_2 N \cdot \mathrm{clog}_3 N$) and matches the verified conditional lower boxes in leading order: with $\beta > 12$ the leading coefficient is below $1/24$.

There is also an unconditional growth reformulation. The conjecture holds if and only if

$$\frac{\#(\mathcal{C} \cap [0, 2^T))}{T^2} \longrightarrow 0.$$

Indeed, under the conjecture the numerator is exactly T; under failure the preceding limit is the strictly positive constant $1/(2\beta)$. This is `alaogluErdosConjecture_iff_candidate_count_below_two_pow_subquadratic`. **[kernel-verified Lean]**

5.5 Block Fourier averages and equidistribution

The fractional part of $x_{N,k}$ is $\{k\beta\}$: a high unit interval contains an initial segment of one irrational rotation orbit.

Theorem 5.9 (Finite Fourier tests on exact blocks)**.** *Assume a counterexample exists and let β be its primitive generator. For every finitely supported coefficient function $c : \mathbb{Z} \to \mathbb{C}$, put*

$$P_c(t) = \sum_{h \in \mathbb{Z}} c(h) e^{2\pi i h t}, \qquad K_N = \Big\lfloor \frac{N+1}{\beta} \Big\rfloor.$$

On the exact ceiling-parametrized points $x_{N,k} = \lceil N - k\beta \rceil + k\beta$,

$$\frac{1}{K_N} \sum_{k=1}^{K_N} P_c(x_{N,k}) \longrightarrow c(0).$$

Status: **[kernel-verified Lean]** — `BlockWeyl` proves vanishing of every nonzero normalized Fourier mode on the exact block points. `BlockEquidistribution` proves `tendsto_primitiveUnitBlockTrigonometricAverage` for every finitely supported coefficient function, with limit `c 0`. This is finite trigonometric-polynomial convergence, not yet a theorem for arbitrary continuous tests.

Theorem 5.10 (Blockwise Weyl equidistribution)**.** *Assume a counterexample exists. For every interval $I \subseteq [0, 1]$ whose boundary has measure zero,*

$$\frac{\#\{x \in (\mathcal{S} \setminus \mathbb{N}_0) \cap [N, N+1) : \{x\} \in I\}}{\#\big((\mathcal{S} \setminus \mathbb{N}_0) \cap [N, N+1)\big)} \longrightarrow |I| \qquad (N \to \infty).$$

Proof. By Theorem 5.8 the relevant fractional parts are $\{\beta\}, \{2\beta\}, \ldots, \{K_N\beta\}$ with $K_N = \lfloor (N+1)/\beta \rfloor \to \infty$. For every nonzero integer h the Weyl sum is a finite geometric series,

$$\frac{1}{K} \sum_{k=1}^{K} e^{2\pi i h k \beta} = \frac{e^{2\pi i h \beta}\big(1 - e^{2\pi i h K \beta}\big)}{K\big(1 - e^{2\pi i h \beta}\big)} \longrightarrow 0,$$

since β is irrational. Weyl's criterion [5] concludes. □

Status: **[paper proof in this report]** — the finite-trigonometric-polynomial input is kernel-verified by Theorem 5.9, but the Stone–Weierstrass/Weyl-criterion extension to all continuous tests and then to interval indicators has not been formalized. No effective discrepancy rate follows without further Diophantine input on β: the geometric-sum proof controls each fixed Fourier mode but not uniformly over many modes.

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6 Equivalent one-number and radical formulations

6.1 Localization at the prime three

Theorem 6.1 (Localization reformulation). *Conjecture 2.1 is false if and only if there exists an odd integer $u > 1$ with*

$$u^{\vartheta} \in \mathbb{Z}[1/3],$$

equivalently, an odd $u > 1$ and $a \in \mathbb{N}_0$ with $3^a u^{\vartheta} \in \mathbb{N}$.

Proof. If x is a counterexample, write $2^x = 2^a u$ with u odd; since 2^x is not a power of two, $u > 1$. Using $(2^a)^{\vartheta} = 3^a$,

$$u^{\vartheta} = \frac{(2^x)^{\vartheta}}{(2^a)^{\vartheta}} = \frac{3^x}{3^a} \in \mathbb{Z}[1/3].$$

Conversely, if $u^{\vartheta} = B/3^a$ with $B \in \mathbb{N}$, set $x = a + \log_2 u$; then $2^x = 2^a u \in \mathbb{N}$ and $3^x = 3^a u^{\vartheta} = B \in \mathbb{N}$, and $x \notin \mathbb{Z}$ since otherwise the odd $u > 1$ would be a power of two. $\square$

Status: [kernel-verified Lean] — `not_alaogluErdosConjecture_iff_exists_odd_localized_radical` and `alaogluErdosConjecture_iff_oddLocalizedRadicalExclusion` in `RadicalReformulation`; an equivalent candidate-level form, `alaogluErdosConjecture_iff_no_odd_multiple_candidate` together with the analytic $\mathbb{Z}[1/3]$-form, is proved independently in `Localization`. The equivalence is kernel-verified; the localized-radical exclusion on its right-hand side remains the open conjecture. The allowance of 3-power denominators is exactly the freedom created by shifting the exponent by an integer.

6.2 Canonical radical algebraicity

In the notation of Theorem 5.3, $E = 3^{\log_2 w}$ satisfies $E^d = A/3^a \in \mathbb{Q}$. Minimality of d has an algebraic consequence.

Proposition 6.2 (Exact radical degree). *The binomial $X^d - A/3^a$ is irreducible over $\mathbb{Q}$, and E is algebraic of exact degree d.*

Proof. For a positive rational q, the binomial $X^d - q$ can be reducible only if q is a p-th power in $\mathbb{Q}$ for some prime $p \mid d$ (the additional $4 \mid d$ exceptional case concerns -4 times a fourth power and is irrelevant for $q > 0$). If $q = r^p$ with $p \mid d$, the positive real $E^{d/p}$ is the positive p-th root of q, hence rational, contradicting minimality of d. $\square$

Status: [kernel-verified Lean] — `irreducible_X_pow_sub_C_of_least_rational_power` in `RadicalDegree` proves the general norm argument, and `oddCoreRpow_exact_radical_degree` specializes it to the least normalized radical index. It identifies the minimal polynomial with $X^d - A/3^a$ and proves its degree is exactly d.

A counterexample therefore produces the very special algebraic number $E = 3^{\log_2 w}$, reached from a transcendental exponent. Taking logarithms of $E^d = A/3^a$:

$$d \log w \log 3 = (\log A - a \log 3) \log 2. \tag{4}$$

This is a *bilinear* relation among logarithms of integers. Baker's theorem controls linear forms in logarithms with algebraic coefficients [3]; it does not rule out (4). The four exponentials conjecture does. The equation is a precise description of the transcendence-theoretic wall, not a disguised elementary factorization problem.

6.3 Where the wall stands in the smallest case

The smallest structural question already meets the open core. A 3-smooth candidate $m = 2^a 3^b$ $(b \geq 1)$ would require $3^a E_3^b \in \mathbb{N}$ for

$$E_3 = 3^{\log_2 3} = e^{(\log 3)^2/\log 2} = 5.7045224946911176\ldots,$$

and even the irrationality of E_3 is open: it is a "three-logarithms" quantity outside the reach of Gelfond–Schneider, Baker's method, and the six exponentials theorem. No currently known technique excludes odd core $w = 3$; this is why the corpus emphasizes counting, structure, and verified finite regions rather than infinite families of exclusions, and why Conjecture 6.3 below is the honest name of the obstruction.

Conjecture 6.3 (Radical four-exponentials target)**.** *For every odd integer $u > 1$, $u^{\log_2 3} \notin \mathbb{Z}[1/3]$. Equivalently, for every primitive odd $w > 1$ and every $d \geq 1$, $(3^{\log_2 w})^d \notin \mathbb{Z}[1/3]$.*

Status: [kernel-verified Lean] — `PrimitiveRadical` proves unique primitive-power decomposition of every integer $u > 1$ and the exact equivalence `alaogluErdosConjecture_iff_primitiveOddLocalizedRadicalExclusion`. Thus proper-power choices of u introduce no genuinely new cases.

By Theorem 6.1 both formulations are exactly equivalent to Conjecture 2.1. A proof of the special case $u = 3$ — even of the weaker assertion that $3^{\log_2 3}$ is irrational — by any genuinely new mechanism would already be a meaningful breakthrough.

7 Sharper finite-difference obstructions

The earlier local theorem (Theorem 4.5) forbids three candidates in progression under the convenient condition $h^5 \leq n$. The sharp arbitrary-order curvature theorem is now kernel-verified for every $r \geq 2$, including the three- and four-term special cases.

For a function f and step $h > 0$ write $\Delta_h f(t) = f(t+h) - f(t)$ and $\Delta_h^r = \Delta_h \circ \cdots \circ \Delta_h$. Iterating the fundamental theorem of calculus,

$$\Delta_h^r f(n) = \int_{[0,h]^r} f^{(r)}(n + t_1 + \cdots + t_r)\, dt_1 \cdots dt_r. \tag{5}$$

For $f(t) = t^\vartheta$ and $r \geq 2$, $f^{(r)}(t) = \vartheta(\vartheta - 1)\cdots(\vartheta - r + 1)\, t^{\vartheta - r}$; since $1 < \vartheta < 2$ this has a fixed nonzero sign on $(0, \infty)$ and its absolute value decreases in t.

Theorem 7.1 (Higher-difference progression obstruction). *Let $r \geq 2$, $n, h \in \mathbb{N}_{>0}$, and $c_r = |\vartheta(\vartheta-1)\cdots(\vartheta-r+1)|$. If all of $n, n+h, \ldots, n+rh$ are candidates, then $1 \leq c_r h^r n^{\vartheta-r}$. Such a progression is therefore impossible whenever*

$$c_r h^r < n^{r-\vartheta}. \tag{6}$$

Proof. If all $n+jh$ are candidates, every $(n+jh)^\vartheta$ is an integer, so the integer combination $\Delta_h^r f(n)$ is an integer; by (5) and the fixed sign of $f^{(r)}$ it is nonzero, hence of absolute value at least one. Bounding $|f^{(r)}|$ by its value at n,

$$1 \leq |\Delta_h^r f(n)| \leq c_r h^r n^{\vartheta-r}.$$

□

Status: [kernel-verified Lean] — `HigherDifference` proves the generic iterated mean-value identity `exists_iterated_fwdDiff_eq_pow_mul`, its real-power specialization `exists_iterated_fwdDiff_rpow_point`, integrality of the binomial forward difference, and `not_arithmetic_progression_twoBaseNaturalCandidates_of_curvature` uniformly for every $r \geq 2$. The displayed iterated-integral identity (5) is not used by the Lean proof; the formal argument instead iterates the mean-value theorem.

For $r = 2$ this gives the sharp three-term criterion

$$\vartheta(\vartheta-1)h^2 < n^{2-\vartheta}, \qquad \text{i.e.} \qquad h < 1.038547802\ldots \cdot n^{0.2075187\ldots}, \tag{7}$$

strictly weaker as a hypothesis than $h \leq n^{1/5}$; this sharp criterion is the theorem `second_difference_logThreeDivLogTwo_ap_of_curvature` in `SharperProgressions`. For $r = 3$, `ThirdDifference` kernel-checks the exact third-difference identity and the corresponding four-candidate obstruction. The first thresholds $h < C_r n^{1-\vartheta/r}$, $C_r = c_r^{-1/r}$:

r	length $r+1$	exponent $1-\vartheta/r$	C_r
2	3	0.207518750	1.038547802
3	4	0.471679166	1.374849673
4	5	0.603759375	1.164124485
5	6	0.683007500	0.946705202
6	7	0.735839583	0.778537835
7	8	0.773576786	0.652646038
8	9	0.801879687	0.557368999

As r grows the admissible step exponent approaches one. This looks tantalizing but does not contradict the exact candidate monoid: in a dyadic block near X the conditional number of candidates is $O(\log X)$, so the average additive gap is of order $X/\log X$, which for each fixed r exceeds $X^{1-\vartheta/r}$. Long-progression leverage would need r growing with the height together with control of the rapidly growing c_r and a combinatorial source of such progressions.

8 Independent finite verification through 2^{20}

Proposition 8.1 (Finite exclusion under interval-software trust). *For every integer $1 \leq m < 2^{20}$: if $m^\vartheta \in \mathbb{Z}$ then m is a power of two. Equivalently, $2^x, 3^x \in \mathbb{Z}$ and $2^x < 2^{20}$ imply $x \in \mathbb{Z}$; a nonintegral solution satisfies $x > 20$.*

Status: [interval computation] — CPython 3.13.5, `mpmath` 1.3.0 interval arithmetic at 40 decimal digits, outward endpoints. Reproducible but *not* replayed by the Lean kernel; not interchangeable with `twelve_le_of_not_integer_of_two_three_rpow_integer`.

For each non-power of two m, a floating approximation proposes $n = \lfloor m^{\vartheta} \rfloor$; the proposal is then *certified* by interval proofs of the two strict inequalities

$$\log m \, \log 3 - \log n \, \log 2 > 0, \tag{8}$$

$$\log(n+1) \, \log 2 - \log m \, \log 3 > 0, \tag{9}$$

which together say exactly $n < m^{\vartheta} < n+1$. Once both interval lower endpoints are positive the floating proposal has no remaining logical role. All 1,048,555 non-powers of two below 2^{20} passed, in two ranges:

Range	Checked	Failures	Smallest certified log margins
$1 \le m < 2^{19}$	524,268	0	lower 1.4147×10^{-15} at $(m, n) = (189427, 231498127)$; upper 3.0841×10^{-15} at $(306982, 497586055)$.
$2^{19} \le m < 2^{20}$	524,287	0	lower 9.4902×10^{-16} at $(958460, 3023921230)$; upper 2.3020×10^{-15} at $(556061, 1275861767)$.

At the globally tightest case the 40-digit interval had width about 1.3×10^{-54}, some thirty-nine orders of magnitude below the certified positive margin; a separate 100-digit computation gives $958460^{\vartheta} - 3023921230 = 4.14020498680340526 \times 10^{-6}$. The verification script, full outputs, and SHA-256 manifests are reproduced in Appendix D.

From scan to kernel certificate. The repository's own architecture (section 4.2) shows the stronger trust model: generate rational certificates externally, revalidate them independently, and replay only exact integer comparisons in Lean by `decide`. A scalable 2^{20} formal certificate should share convergent enclosures across ranges, chunk the data modules to bound elaboration memory, and rely on verified binary powering rather than giant literals; the worst cases identified by the scan (the four extremal pairs above) predict the required tier depth.

9 Attempts at a full proof and their exact failure points

9.1 Collapse rank two to rank one

The rank argument leaves a perfectly coherent rank-two possibility, now known exactly: $\mathcal{C} = \langle 2, M \rangle$. Nothing in the rank machinery distinguishes this from the allowed monoid of powers of two; excluding the second generator *is* the original transcendence problem.

9.2 Density of multiples against endpoint gaps

At paper level the orbit $k\beta$ mod 1 is equidistributed (Theorem 5.10); the finite-Fourier convergence needed for Weyl's criterion is kernel-verified by Theorem 5.9. The integrality gap at height k is of exponential scale M^{-k} (Theorem 4.2). Equidistribution controls fixed intervals — and with effective discrepancy, polynomially shrinking ones along subsequences — but numbers with bounded partial quotients show that equidistributed multiples can avoid exponentially small targets forever. Closing this route needs Diophantine information specific to the transcendental β with $2^\beta, 3^\beta \in \mathbb{N}$, not generic metric theory.

9.3 Exact counts against local progression obstructions

Conditional failure gives $O(\log X)$ candidates per multiplicative block — far below the positive-density regime of Roth-type theorems — while Theorem 7.1 reaches steps $X^{1-\vartheta/r}$ against average gaps $X/\log X$. Sparse exact structure and local curvature are mutually consistent. A three-term progression of candidates would solve the S-unit equation $2^{n_1}M^{k_1} + 2^{n_3}M^{k_3} = 2^{n_2+1}M^{k_2}$; general S-unit theory yields finiteness statements, not the required nonexistence.

9.4 Exploit the primitive output pair

The pair (M, A) admits no matched depth, no common perfect power, and no nontrivial affine decomposition (Corollaries 5.4, 5.5 and 5.6). What remains is exactly relation (4), a product of logarithms outside the scope of linear-forms machinery.

9.5 Force algebraicity of a forbidden auxiliary exponential

The five exponentials theorem [10, 6] makes $e^{\gamma x}$ transcendental for every nonzero algebraic γ, given the four algebraic corner exponentials. The natural auxiliary quantities $p^x = e^{x \log p}$ have transcendental coefficient $\log p$, so the theorem does not directly convert divisors of M or A into a contradiction. The six exponentials theorem shows b^x is transcendental for any base b multiplicatively independent of $2, 3$; the committed Theorem 4.13 is its formalized integer-level shadow. Neither prevents b from dividing the integer 2^x; they control the distinct quantity b^x.

9.6 The linear-form-in-log-products formulation of the wall

The obstruction can be given one further exact name. A counterexample is a pair of integers $m, A \geq 2$ with $\log m \log 3 = \log A \log 2$. Expanding both integers over their prime factorizations $m = \prod_p p^{m_p}$, $A = \prod_p p^{a_p}$, failure of the conjecture is equivalent to the nontrivial vanishing of

$$D \;=\; \sum_p \bigl(m_p \log 3 - a_p \log 2\bigr) \log p,$$

an *integer-coefficient linear form in the products of two prime logarithms* $\{\log p \cdot \log 2,\ \log p \cdot \log 3\}$. Baker's theory bounds linear forms in logarithms with *algebraic* coefficients; here every coefficient of $\log p$ is itself a transcendental logarithm, and no lower-bound technology exists for such forms. Indeed even the $\mathbb{Q}$-linear independence of the family $\{\log p \cdot \log q : p < q \text{ prime}\}$ is an open problem (it follows from, and is essentially a fragment of, the four exponentials circle of conjectures). Every route examined in this report — determinants, product formulas in the radical field $\mathbb{Q}(E)$, auxiliary exponentials, equidistribution against shrinking targets — terminates at a special case of exactly this independence statement. A genuine proof therefore requires either a fundamentally new lower bound for linear forms in log-products, or a mechanism exploiting the positivity and integrality of the coefficients (m_p, a_p) that has no analogue in the general four-exponentials setting.

10 Consolidated research program

The two source documents proposed overlapping programs; the exact structure theory settles some directions and re-prioritizes the rest. Ordered by leverage and feasibility:

Priority A: finish the exact conditional consequences

The primitive normalization, uniqueness of the odd core and its coordinates, least non-integer solution and affine primitivity, simultaneous two-output normalization, rational-power index, denominator support, and exact primitive-generator theorem are complete in `PrimitiveOddCore`, `LeastSolution`, `OutputNormalization`, `ThreeDenominatorNormalization`, `RationalPowerIndex`, and `PrimitiveGenerator`. `ExactCounting` and `QuadraticAsymptotic` prove the exact unit-block and dyadic counts, the cumulative finite sum, its sharp quadratic limit, and the subquadratic-growth equivalence. `BlockWeyl` and `BlockEquidistribution` prove every fixed Fourier mode and finite trigonometric polynomial. The remaining work in this priority is the functional-analytic extension to arbitrary continuous tests and interval indicators in Theorem 5.10. This supersedes the earlier "exponent semigroup" direction: the kernel-verified classification shows there are no Frobenius-type gaps to study.

Priority B: exploit the uniform higher-difference theorem

`HigherDifference` now formalizes a generic iterated forward-difference mean-value identity and proves Theorem 7.1 uniformly for every $r \geq 2$. The remaining problem is no longer analytic formalization but combinatorial leverage: propagate the sharp variable-order criterion through sparse-set arguments, with r allowed to grow with height, and determine whether the rapidly growing falling coefficient still permits a contradiction with the exact conditional monoid. For concrete finite verification, the rational effective three-term criterion $h^{400} \leq n^{83}$ (`not_three_arithmetic_progression_twoBaseNaturalCandidates_sharp` in `SharpProgression`, with the kernel-replayed enclosure $\vartheta < 317/200$ from $3^{200} < 2^{317}$) complements the exact-curvature form.

Priority C: kernel-replay the 2^{20} region

The scan shows the finite truth is computationally routine; the bottleneck is a compact kernel-replayable proof object. Adaptive convergent enclosures, chunked certificate modules, and shared tiers along ranges are the ingredients; the target theorem is the 1048576 analogue of Theorem 4.3. Every descent constant then improves to 20 by bookkeeping.

Priority D: use the exact radical field

`RadicalReformulation` now formalizes Theorem 6.1 and its symmetric base-swapped version, `PrimitiveRadical` reduces it canonically to power-primitive odd bases, while `PrimitiveGenerator` and `ThreeDenominatorNormalization` supply the primitive odd w, least index d, and reduced $q \in \mathbb{Z}[1/3]$ with $(3^{\log_2 w})^d = q$. `RadicalDegree` proves Proposition 6.2, including the exact minimal polynomial. The next target is a genuinely new number-field argument using this explicit extension; the radical exclusion itself remains equivalent to, and exactly as open as, the original conjecture.

Priority E: five and sharp six exponentials

Formalizing the five exponentials theorem or Roy's sharp six exponentials theorem [6] would add restrictions such as transcendence of $e^{\gamma x}$ for algebraic $\gamma \neq 0$. The effort is comparable to the completed Gelfond–Schneider port and should be undertaken with a specific downstream lemma in mind. `RationalThirdBase` already handles a rational third output when its reduced denominator divides a product of powers of the two integral outputs, and proves a determinant lower bound and an explicit denominator margin. Extending the interpolation grid to an unrestricted rational output remains open.

Priority F: auxiliary-function methods tailored to integrality

A route beyond four exponentials must exploit more than algebraicity: here all corner values are positive integers, all their powers remain integers, and the primitive pair is canonical. Promising variants: use several non-Archimedean valuations simultaneously; use Corollary 5.6 to prevent determinant factors from sharing perfect powers; work with norms in the explicit radical field $\mathbb{Q}(E)$ of Proposition 6.2, combining Archimedean smallness with denominator support only at 3; search for a product-formula contradiction across all places of $\mathbb{Q}(E)$. The last is the most structurally aligned with the radical formulation.

Priority G: primitive-pair search instead of raw candidate scanning

A counterexample has a primitive pair (M, A) with $M^\vartheta = A$, $\min(v_2 M, v_3 A) = 0$, no common perfect power, and no matched affine decomposition. Finite searches can reject candidates early on these conditions and record near misses keyed by odd core, revealing whether closest approaches cluster by smoothness pattern, denominator, or rational-power index d.

Priority H: quantitative irrationality and discrepancy

Effective irrationality measures for ϑ (Páde/hypergeometric methods) would replace per-value certificate tiers by a uniform polynomial criterion and likely extend verified regions by orders of magnitude. They do not directly control the generator $\beta = \log_2 M$ for unknown M; that would require an effective measure derived from the relation (4), which lies beyond ordinary linear forms in logarithms.

Downstream: the colossally abundant interface

Formalizing the original application — the conjecture's rational-power form implies that quotients of consecutive colossally abundant numbers are prime [1] — connects the corpus to its historical source. It is logically downstream and should follow the core structure modules.

11 Conclusions

The repository contains a broad, kernel-verified attack on the Alaoglu–Erdős conjecture whose themes are complementary: interpolation determinants settle three independent bases; Gelfond–Schneider excludes algebraic irrational exponents; exact certificates create a growing zero-free region; unique factorization yields a canonical primitive odd core and an exact conditional solution monoid; arbitrary-order forward differences force sharp local additive sparsity; leastness gives exact affine primitivity; and unconditional equivalences with fractional-part gaps and localized radicals identify the entire remaining distance to the conjecture.

The conjecture does not fall to a combination of these results, and the reason can now be stated exactly. A counterexample would not generate a messy exceptional set that density or semigroup arguments might squeeze; it would generate the perfectly rigid free monoid

$$\mathcal{S} = \mathbb{N}_0 + \mathbb{N}_0\beta, \qquad \mathcal{C} = \{2^n M^k : n, k \in \mathbb{N}_0\},$$

whose block counts, fractional parts, and multiplicative relations are explicit and internally consistent with every verified gap and density bound. The remaining assertion is that the second generator cannot exist — equivalently (Conjecture 6.3), that no odd $u > 1$ has $u^{\log_2 3} \in \mathbb{Z}[1/3]$.

The free-monoid classification, simultaneous primitive output normalization, exact block counts, sharp cumulative asymptotic, finite-trigonometric-polynomial block convergence, primitive localized-radical reduction, exact radical degree, and uniform higher-difference obstruction are kernel-verified. Full blockwise equidistribution for arbitrary continuous tests and intervals remains a paper theorem, as does any unrestricted rational-third-output determinant grid.

That reformulation is progress rather than pessimism: it removes artificial degrees of freedom, supplies a canonical primitive object, yields exact conditional counting statements, and gives a focused interface for future transcendence methods. The near-term program is listed in section 10; a complete proof still requires a new idea at the four-exponentials boundary, but the target is now narrow and the surrounding formal infrastructure is strong enough to test such an idea precisely.

A Status matrix for major statements

Statement	Status	Location
$2^x, 3^x, 5^x \in \mathbb{Z} \Rightarrow x \in \mathbb{Z}$	**[kernel-verified Lean]**	`IntegerExponent`
Nonintegral solution irrational, transcendental	**[kernel-verified Lean]**	`TwoBaseIntegerExponent`, `Transcendence`
$2^x < 256 \Rightarrow x \in \mathbb{Z}$	**[kernel-verified Lean]**	`FiniteCheck256`

Statement	Status	Location
$2^x < 4096 \Rightarrow x \in \mathbb{Z}$ $(x \geq 12)$	[kernel-verified Lean]	`FiniteCheck4096`
Rank of candidate exponent vectors ≤ 2	[kernel-verified Lean]	`MultiplicativeRank`
Common odd core $2^i w^j$ (existence)	[kernel-verified Lean]	`OddCore`
Counting $\operatorname{clog}_2 N \cdot \operatorname{clog}_3 N$; linear dyadic blocks	[kernel-verified Lean]	`OddCore`
Density dichotomy; gap equivalences; infinitude	[kernel-verified Lean]	`FractionalPartDensity`
3-smooth classification of further bases	[kernel-verified Lean]	`ThreeSmooth`
Affine descent, $r + 8k \leq x$	[kernel-verified Lean]	`ArithmeticRigidity`
Three-term AP obstruction under $h^5 \leq n$	[kernel-verified Lean]	`ZeroDensity`
Shrinking-target Diophantine bounds	[kernel-verified Lean]	`ShrinkingTarget`
Gelfond–Schneider theorem	[kernel-verified Lean]	`GelfondSchneider`
4EC $\Rightarrow$ conjecture	[kernel-verified Lean]	`FourExponentialsReduction`
Primitive (canonical) odd core; unique coordinates	[kernel-verified Lean]	`PrimitiveOddCore`
Unique least solution; affine primitive output pair	[kernel-verified Lean]	`LeastSolution`
Simultaneous primitive output normalization; gcd one; size dichotomy	[kernel-verified Lean]	`OutputNormalization`
Power-of-three denominator normalization	[kernel-verified Lean]	`ThreeDenominatorNormalization`
Rational-power index and exact solution monoid $\mathbb{N}_0 + \mathbb{N}_0\beta$	[kernel-verified Lean]	`RationalPowerIndex`, `PrimitiveGenerator`
Exact block counts and cumulative finite sum	[kernel-verified Lean]	`ExactCounting`
Cumulative $T^2/(2\beta) + O(T)$ asymptotic and subquadratic iff	[kernel-verified Lean]	`QuadraticAsymptotic`
Exact block averages for finite trigonometric polynomials	[kernel-verified Lean]	`BlockWeyl`, `BlockEquidistribution`
Full continuous-test/interval blockwise equidistribution	[paper proof in this report]	Theorem 5.10
Denominator-controlled rational third output and determinant margin	[kernel-verified Lean]	`RationalThirdBase`
Unrestricted rational-third-output determinant grid	open	not yet formalized
Sharp three-term and four-term AP obstructions $(r = 2, 3)$	[kernel-verified Lean]	`SharperProgressions`, `ThirdDifference`
Uniform higher-order AP obstruction $(r \geq 2)$	[kernel-verified Lean]	`HigherDifference`
Localized-radical and symmetric reformulations	[kernel-verified Lean]	`RadicalReformulation`
Primitive localized-radical reduction	[kernel-verified Lean]	`PrimitiveRadical`

Statement	Status	Location
Exact radical degree and binomial irreducibility	**[kernel-verified Lean]**	`RadicalDegree`
Effective rational three-term criterion ($h^{400} \le n^{83}$)	**[kernel-verified Lean]**	`SharpProgression`
Candidate-form localization equivalence	**[kernel-verified Lean]**	`Localization`
Convergent-denominator growth $q_{n+1} < 2m^{q_n}$	**[kernel-verified Lean]**	`DenominatorGrowth`
No nonintegral candidate with $2^x < 2^{20}$	**[interval computation]**	Proposition 8.1
Full Alaoglu–Erdős conjecture	open	reduced to Conjecture 6.3

B Formalization inventory

All modules live under `Analysis/ExponentialIdentities/Lean/ExponentialIdentities/`, in namespace `LeanProofs`. No `sorry`, no extra axioms, no `native_decide`; certificate tables are replayed by the kernel.

Module	Contents
`IntegerExponent` (+3 modules)	three-base theorem, interpolation determinants
`TwoBaseIntegerExponent`	conjecture statement, candidates, gaps, $2^x < 10$
`.../FiniteCheck`, `FiniteCheck256`	zero-free regions $2^x < 100$, $2^x < 256$
`.../FiniteCheck4096`	zero-free region $2^x < 4096$ (kernel `decide` table)
`.../ZeroDensity`	AP obstruction, Roth-number counting, $o(N)$
`.../NonintegerDensity`	nonintegral candidate refinement, density zero
`.../MultiplicativeRank`	rank ≤ 2, ϑ irrational, $(c\log_2 N)^2$
`.../CandidateStructure`	cross-valuation identity, conditional box growth
`.../OddCore`	common odd core, $c\log_2 N \, c\log_3 N$, dyadic blocks
`.../PrimitiveOddCore`	gcd-normalized canonical core, unique coordinates
`.../LeastSolution`	unique positive least solution, exact affine primitivity
`.../ThreeDenominatorNormalization`	reduced denominator supported at 3
`.../RationalPowerIndex`	least rational-power index, exact solution monoid
`.../PrimitiveGenerator`	canonical primitive generator and unique coordinates
`.../ExactCounting`	exact unit/dyadic block counts and cumulative finite sum
`.../QuadraticAsymptotic`	sharp quadratic limit; subquadratic-growth equivalence
`.../BlockWeyl`	exact-block geometric sums and vanishing nonzero Fourier modes
`.../BlockEquidistribution`	finite trigonometric-polynomial block averages
`.../OutputNormalization`	simultaneous primitive outputs, gcd one, size dichotomy
`.../RationalThirdBase`	denominator divisibility and determinant margin
`.../RadicalReformulation`	localized-radical equivalences, base-swapped form
`.../PrimitiveRadical`	canonical primitive-power reduction of radical target
`.../RadicalDegree`	irreducible binomial, exact minpoly and degree
`.../SharperProgressions`	sharp three-candidate curvature obstruction
`.../ThirdDifference`	exact third difference, four-candidate obstruction
`.../HigherDifference`	uniform arbitrary-order forward-difference obstruction
`.../ArithmeticRigidity`	descent, $r + 8k \le x$
`.../ThreeSmooth`	3-smooth classification of further bases
`.../ShrinkingTarget`	fract shrinking targets, Diophantine lower bounds
`.../DenominatorGrowth`	convergent-denominator growth cap $Q < 2m^q$
`.../Transcendence`	GS applications: transcendence of solutions, of $\log_2 3$
`.../FractionalPartDensity`	density dichotomy, gap equivalences, infinitude
`.../Localization`	candidate-form localization equivalence
`.../SharpProgression`	effective rational three-term criterion
`.../FourExponentialsReduction`	4EC $\Rightarrow$ conjecture
`GelfondSchneider/...`	Gelfond–Schneider theorem (port of Karatarakis)

C A symmetric formulation

Everything above has a base-swapped version. Put $\eta = \log_3 2 = 1/\vartheta$. Failure of the conjecture is also equivalent to the existence of an integer $v > 1$ with $3 \nmid v$ and

$$v^\eta \in \mathbb{Z}[1/2].$$

Status: [kernel-verified Lean] — `not_alaogluErdosConjecture_iff_exists_threeFree_localized_radical` and `alaogluErdosConjecture_iff_threeFreeLocalizedRadicalExclusion` in `RadicalReformulation`.

For the primitive pair (M, A), factoring A into a power of 3 times a 3-coprime part produces a symmetric primitive core and rational-power index; the two normalizations give the same least exponent β, and their compatibility condition is exactly $\log M \log 3 = \log A \log 2$. The symmetry is useful in formalization — some divisibility arguments are easier on one side — and confirms that Corollary 5.4 is not an artifact of privileging the base 2.

D Interval computation: script and outputs

```python
import math, time, hashlib
import mpmath as mp

START = 1
LIMIT = 1 << 19
DPS = 40

mp.iv.dps = DPS
ln2 = mp.iv.log(2)
ln3 = mp.iv.log(3)
theta_float = math.log(3.0) / math.log(2.0)

checked = 0
failures = []
min_lower = (float("inf"), None, None)
min_upper = (float("inf"), None, None)
manifest = hashlib.sha256()
start_time = time.time()

for m in range(START, LIMIT):
    if (m & (m - 1)) == 0:      # integral exponent: m is a power of two
        continue

    # Only proposes a neighboring integer.  The interval tests below certify it.
    n = math.floor(math.exp(theta_float * math.log(m)))

    lhs = mp.iv.log(m) * ln3
    lower_diff = lhs - mp.iv.log(n) * ln2
    upper_diff = mp.iv.log(n + 1) * ln2 - lhs

    lower_lb = float(lower_diff.a)
    upper_lb = float(upper_diff.a)
    if not (lower_lb > 0.0 and upper_lb > 0.0):
        failures.append((m, n, str(lower_diff), str(upper_diff)))
        break

    min_lower = min(min_lower, (lower_lb, m, n))
    min_upper = min(min_upper, (upper_lb, m, n))
    manifest.update(f"{m}:{n};".encode())
    checked += 1

print("checked_nonpowers=", checked)
print("failures=", len(failures))
print("min_lower_log_margin=", min_lower)
print("min_upper_log_margin=", min_upper)
print("sha256_m_n_manifest=", manifest.hexdigest())
print("elapsed_seconds=", time.time() - start_time)
assert not failures
```

Listing 1: Interval verification for one range. Set START to 1 or 2^{19} and LIMIT to 2^{19} or 2^{20}.

The two 40-digit runs produced:

```
LIMIT=524288
DPS=40
checked_nonpowers=524268
failures=0
min_lower_log_margin=(1.4147089634746718e-15, 189427, 231498127)
min_upper_log_margin=(3.0841010256409534e-15, 306982, 497586055)
sha256_m_n_manifest=02b589461fa8c1721bc251575b98b1e3d58001f54cb7faa2cd6c59edb8c819a5
elapsed_seconds=18.505
```

Listing 2: Lower half output.

```
LIMIT=1048576
DPS=40
checked_nonpowers=524287
failures=0
min_lower_log_margin=(9.490232037370244e-16, 958460, 3023921230)
min_upper_log_margin=(2.3019781003173366e-15, 556061, 1275861767)
sha256_m_n_manifest=fccccf5de8673379d37ff0fc840ab091d9ff7a2352ff64a93091f01c21bf4cb4
elapsed_seconds=18.904
```

Listing 3: Upper half output.

Elapsed times are machine-dependent sanity checks; the counts, zero-failure results, extremal tuples, and hashes are the reproducibility data. The hashes cover the ASCII stream `m:n;` of proposed-and-certified pairs and make accidental changes to the candidate sequence detectable across reruns.

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